BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings illustrate example embodiments of the systems and methods described herein and are not intended to limit the scope of the disclosure.

FIG. 1 illustrates a perspective view of an example penile geometry scanning device including a housing structure and representative locations for at least one time-of-flight (ToF) ranging sensor and at least one optical motion sensing component.

FIG. 2 illustrates an example measurement configuration in which radial distance samples are captured relative to a penile anatomical surface while the device is displaced along a longitudinal axis, with motion tracking used to associate distance measurements with positional data.

FIG. 3 illustrates a representative sensor layout showing example placement of a ToF ranging module, an optical motion sensor module configured to detect longitudinal displacement, and an optional inertial measurement unit (IMU) configured to compensate for tilt or angular deviation during scanning.

FIG. 4 illustrates a schematic block diagram of an example processing architecture including a ToF acquisition subsystem, a motion sensing subsystem, a reconstruction engine configured to generate a three-dimensional geometric model, and a geometry profile generation module configured to output derived structural metrics and/or a standardized digital geometry representation.

It will be understood that the illustrated embodiments are exemplary and that alternative housing configurations, sensor arrangements, sensing modalities, reconstruction algorithms, and integration architectures may be utilized without departing from the spirit and scope of the present disclosure.